

Brendan Tomoschuk

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Experience

PhD Researcher

2014-Present

- Published methods paper modeling language proficiency of bilinguals to 91% accuracy with survey data collected from over 1500 people
- Published NLP paper on predicting errors in early second language learners to 87% accuracy
- Led three-person team to design and test novel word learning methods that improves second language word learning by up to 11% (publication in preparation)
- Co-founded weekly data science learning group focused on peer-to-peer instruction between graduate students in multiple departments

Consulting Intern – Intelliphonics

2018

- Analyzed client data to determine effective assessment questions for non-native speakers
- Designed and coded online experiments evaluating novel language assessment techniques
- Designed A/B tests to determine the effectiveness of mobile language assessment

Fulbright Researcher

2016-2017

- Led collaboration between two labs in San Diego, CA and Ghent, Belgium
- Found significant differences in language learning strategies between US and Belgian populations that inform language theory and pedagogy (publication in preparation)
- Presented findings to the acting American ambassador to Belgium

Teaching Assistant

2014-Present

- Directed weekly statistics and research methods classes to 30 undergraduate students
- Conducted quarterly statistics or cognitive psychology lectures to classes of 300 students

Data Science Projects

Reddit Sentiment Analysis <https://goo.gl/Lz85G4> – Implemented NLP pipeline in Python to scrape, process and analyze topic and sentiment from Reddit comments

Language Learning Difficulty <https://goo.gl/ipnN9S> – Evaluated learning difficulty of different languages from Duolingo user data with R

Transcript Explorer crtranscriptexplorer.herokuapp.com – Built an interactive dashboard in Python that allows users to explore and graph the language use by reading and processing subtitle files

Skills

Proficient

R (tidyverse, ggplot2, lme4), Python (pandas, NumPy, nltk, scikit-learn)

Familiar

HTML, CSS, JavaScript, SQL (PostgreSQL)

Education

Ph.D. Experimental Psychology, *University of California, San Diego*

2014-2019

B. Phil. Neurolinguistics, B.S. Science, *Pennsylvania State University*

2010-2014

Awards & Funding

- Fulbright Fellowship 2016-2017
- National Defense Science and Engineering Graduate (NDSEG) Fellowship 2015-2019
- National Science Foundation (NSF) PIRE Fellowship 2013